

Precision Modeling and Mitigation Strategies for IR Drop, Output Scaling, and Sneak-Path Calibration in ReRAM In-Memory Computing

Introduction and Architectural Paradigms

Analog In-Memory Computing (AIMC) architectures leveraging Resistive Random-Access Memory (ReRAM) crossbar arrays have emerged as a dominant hardware paradigm to overcome the von Neumann memory wall in artificial intelligence workloads¹. By mapping weight matrices to the physical conductance states (G) of memristive devices and applying activation inputs as voltage signals (V), matrix-vector multiplication (MVM) operations can be executed directly within the memory array in $O(1)$ temporal complexity¹. This physical execution relies on Ohm's Law ($I = V \cdot G$) to compute local inner products and Kirchhoff's Current Law ($I_{total} = \sum I_i$) to aggregate cumulative currents along array bitlines². Despite these computational density and energy efficiency advantages, scaling ReRAM crossbar arrays from small research testbeds to dense, high-throughput accelerators introduces severe signal integrity degradation⁷. Hardware non-idealities severely distort voltage distribution, total output current, and logical state evaluation across the crossbar grid⁵. Interconnect line parasitic resistances induce significant spatial voltage drops (IR drop) along wordlines and bitlines⁵. Concurrently, device-level current-voltage (I - V) non-linearity causes memristive conductance to fluctuate as a function of the applied terminal voltage⁹. Furthermore, conductance-dependent output scaling bottlenecks arise due to restricted device programming precision, high bitline current summation, and mixed-signal peripheral constraints². In passive crossbar configurations, unintended leakage currents traversing unselected nodes—commonly designated as the sneak-path problem—further degrade operational reliability¹². When unaddressed, these aggregated physical non-idealities cause deep neural network (DNN) classification accuracy to plummet—for instance, dropping from an ideal floating-point accuracy of 97% down to 59% in uncompensated crossbars⁷. Amortizing the overhead of peripheral Analog-to-Digital Converters (ADCs) and Digital-to-Analog Converters (DACs) requires large crossbar tiles (128×128 up to 2048×2048), which simultaneously amplifies the impact of parasitic parameters¹. Developing rapid, highly accurate computational models alongside circuit-level and algorithmic compensation frameworks is therefore mandatory for functional

AIMC deployment⁵.

Mathematical Modeling and Analysis of Parasitic IR Drop

Physical Mechanisms and Non-Linear Cyclic Coupling

In an ideal ReRAM crossbar, wordline driver voltages propagate without attenuation to every crosspoint cell, and bitline currents flow directly into sense amplifiers at zero potential. However, real metallic interconnect lines possess finite parasitic resistances (r_{par} , typically ranging from 5Ω to 20Ω per segment)⁵. In a representative crossbar schematic, nominal input voltages applied at the wordline drivers traverse sequential wire segments characterized by parasitic resistance r_{par} to reach internal node $V_{WL}(i, j)$. Current flows through memristive element $G(i, j)$ down to bitline node $V_{BL}(i, j)$, which subsequently traverses successive column line parasitic resistance segments before reaching the output sensing transimpedance amplifier⁷. As current flows through this resistive network, voltage drops accumulate along both rows and columns⁵. Consequently, a memristive device situated at row i and column j experiences an effective differential voltage $V_{cell}(i, j) = V_{WL}(i, j) - V_{BL}(i, j)$ that is substantially lower than the nominal input voltage $V_{in}(i)$ ⁸.

This phenomenon exhibits a complex cyclic dependency when coupled with device $I-V$ non-linearity⁹. Because ReRAM devices exhibit voltage-dependent conductance $G(V_{cell})$, the voltage drop $\Delta V_{IR}(i, j)$ across the parasitic wire segments dictates the localized device conductance $G(i, j)$ ⁹. Reciprocally, the localized device conductance determines the magnitude of current drawn from the wordline into the bitline, which directly re-modifies the voltage drop across all preceding and succeeding wire segments in the array⁹. This spatial coupling destroys the linearity of the matrix-vector multiplication, causing the array output current to deviate non-linearly from the true mathematical dot-product⁴.

Computational Modeling Paradigms

Quantifying the impact of IR drop across large-scale crossbars requires solving a system of nodal voltage equations derived from Kirchhoff's laws⁷. For an $M \times N$ crossbar, the nodal analysis yields a system of linear or non-linear equations represented as:

$$\mathbf{M} \cdot \mathbf{V} = \mathbf{I}$$

where $\mathbf{M}$ is the circuit conductance matrix incorporating both parasitic wire conductances $g_{par} = 1/r_{par}$ and cell conductances $G_{i,j}$, $\mathbf{V}$ is the node voltage vector, and $\mathbf{I}$ represents the external boundary current injections⁷. Three primary modeling paradigms exist to solve this formulation.

Compact analytical models employ closed-form spatial integrals or homogenized continuous approximations to estimate average array voltage degradation⁵. While computationally trivial ($O(1)$ to $O(N)$ execution time), analytical models assume uniform weight distributions and ignore localized current clustering⁵. They retain reasonable fidelity only for small array

dimensions ($\leq 32 \times 32$) and fail under arbitrary, sparse, or non-linear weight patterns⁵.

Standard SPICE simulators perform full transient or DC non-linear nodal analysis⁷. While providing maximum physical accuracy, SPICE computation scales at $O(N^3)$ to $O(N^4)$

with respect to total node count⁷. Simulating medium-to-large arrays (256×256 up to 2048×2048) within machine learning training loops becomes computationally intractable, requiring hours or days for a single forward pass¹.

Iterative refinement and numerical matrix solvers utilize specialized numerical linear algebra algorithms designed specifically for crossbar topographies⁷. Iterative refinement algorithms separate linear wire parasitic effects from non-linear device conductances, solving a fixed-point equation system:

$$\mathbf{V}^{(k+1)} = \mathcal{F}(\mathbf{V}^{(k)}, G(\mathbf{V}^{(k)}))$$

This process converges rapidly within 4 to 6 iterations, reducing root-mean-square (RMS) output error by over $10\times$ compared to decoupled linear models⁹. Alternatively, super-nodal sparse LU factorization and Anderson acceleration decouple linear parasitic networks into optimized sparse matrices, achieving $500\times$ to $1250\times$ execution speedups over SPICE while preserving numerical accuracy within 10^{-15} ⁸.

Modeling Methodology	Time Complexity	Speedup vs. SPICE	Numerical Precision / RMS Error	Handling of Device Non-Linearity
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Compact Analytical Models	$O(1) -$	$>$	Low ($>$ error for arrays $> 32 \times$)	Linearized / Neglected ⁷
SPICE / HSPICE	$O(N^3) -$	$1 \times$ (Baseline)	Exact Reference (10^{-15})	Full Non-Linear Transistor/Memristor Models ⁷
Iterative Refinement	$O(k \cdot$	$\sim 100 \times$	High ($<$ error, $10 \times$ RMS reduction over analytical)	Solved via Fixed-Point Iteration ¹
Super-Nodal Sparse LU + Anderson	$O(N \cdot$	$506.8 \times$	Ultra-High (10^{-15} precision match to SPICE)	Equivalent Linear Transformation / Accelerated Fixed-Point ⁸

Algorithmic and Architectural Compensation for IR Drop

Mitigating the loss of inference accuracy due to IR drop requires combining software-level retraining algorithms, weight mapping strategies, and specialized circuit architectures¹.

Software and Algorithmic Compensation Strategies

Standard deep learning training maps floating-point weights directly to memristive conductances without considering voltage drop⁴. Offline Non-Ideality Aware Training (NAT) incorporates predicted IR drop distortions directly into the forward pass of model training¹. However, calculating data-dependent IR drop for every mini-batch via conventional simulation severely degrades training throughput¹.

To resolve this bottleneck, the Weight-centric Prediction Model (WPM) and Stochastic WPM (S-WPM) translate the effect of IR drop and device non-linearity from an output-dependent noise term into a static weight matrix distortion⁴:

$$W_{effective} = \mathcal{G}_{distortion}(W, \mathbf{x}_{random})$$

Whereas standard inference passes inputs through ideal weights to produce hardware-degraded output currents, the WPM framework transforms the target weights into an effective pre-distorted matrix $W_{effective}$ prior to hardware execution⁴. This transformation causes the physical crossbar output current under hardware non-idealities to match the expected ideal mathematical result⁴. By utilizing tailored input randomization, data-dependency is eliminated during training iterations¹. This enables the generation of distorted weight matrices that can be reused across training epochs, accelerating offline NAT by $100\times$ to $15,000\times$ compared to SPICE-in-the-loop approaches while reducing prediction RMS error by $2\times$ to $90\times$.

For pre-trained Convolutional Neural Networks (CNNs), performing full network retraining may be computationally expensive or restricted by dataset availability¹. Retraining-free compensation techniques exploit the linear properties of Batch Normalization (BN) layers following crossbar MVM engines¹. Because IR drop primarily induces a systematic attenuation and spatial shift of column currents, the running mean (μ_{BN}) and running variance (σ_{BN}^2) parameters of adjacent BN layers can be recalculated post-hoc using a small validation subset¹. Adjusting the BN scale (γ) and shift (β) parameters directly absorbs the average current deficit, yielding up to a 35.55% recovery in classification accuracy and bringing hardware performance within 0.16% of ideal floating-point baselines¹.

Circuit-Level and Architectural Interventions

At the circuit level, dedicated peripheral compensation circuits can be designed to apply dynamic current injection along bitlines⁵. By modeling the linear relationship between total column current and parasitic drop, local current cancellation circuits inject compensating currents directly at the sensing nodes⁵. Board-level hardware experiments demonstrate that Design Technology Co-Optimization (DTCO) based peripheral compensation reduces current distortion from 71% down to 2%, restoring classification accuracy on degraded crossbars from 89% to 95%⁵.

Because the total IR drop voltage magnitude $\Delta V = \sum I_{cell} \cdot r_{par}$ is directly proportional to the total current flowing through the array, reducing active current drastically mitigates parasitic drops⁷. The BinSparX technique applies static sparsification to network weights and dynamic sparsification to input activations in Binarized Neural Networks (BNNs)⁷. By systematically minimizing the density of $+1$ states (mapping them to 0 or -1 representation), total crossbar current draw is suppressed, eliminating severe voltage degradation without requiring hardware modifications or fine-grained retraining⁷. Conventional 1T1R or 1R arrays accumulate total bitline current monotonically¹⁶. The Sign-Weighted 2T2R (SW-2T2R) architecture utilizes two differential ReRAM cells per weight,

combined with sign-dependent activation control¹⁶. If positive and negative differential conductances are closely matched ($G^+ \approx G^-$), the net differential source-line current $I_{SL} = I^+ - I^-$ approaches zero¹⁶. This common-mode current reduction minimizes cumulative voltage drops across array lines, boosting parallel computation capacity while achieving 78.4 TOPS/W peak energy efficiency on fully integrated neuromorphic chips¹⁶.

Compensation Strategy	Implementation Level	Target Mechanism	Accuracy Impact (Uncompensated → Compensated)	Hardware / Overhead Trade-offs
Weight-centric Prediction (WPM)	Algorithmic / NAT	Pre-distorts target weight matrix during offline training	Restores baseline accuracy across complex CNNs (e.g., CIFAR-10, ImageNet) ¹	Zero runtime hardware area/power overhead; requires offline retraining ¹
BN Layer Recalibration	Software Post-Processing	Adjusts BN scale (γ) and shift (β) parameters to absorb current attenuation	Restores accuracy up to +35.55% (within 0.16% of ideal) ¹	Retraining-free ; requires brief recalibration on a small validation dataset ¹
DTCO Peripheral Circuitry	Mixed-Signal Circuit	Injects local cancellation currents along bitline sense amplifiers	Reduces current distortion from 71% to 2% (Accuracy: 89% →) ⁵	Requires peripheral silicon area for operational amplifiers / current mirrors ⁵

BinSparX Sparsification	Algorithmic / Architectural	Reduces density of active high-conductance states (+1)	Restores accuracy from 59% to 96.6% on BNNs ⁷	Training-free; requires activation/weight sparsification logic ⁷
Sign-Weighted 2T2R Array	Cell / Array Topology	Cancels common-mode current via differential sensing ($I^+ -$)	Sustains 94.4% accuracy on MNIST under high parallelism ¹⁶	$2\times$ memristor cell area overhead; yields 78.4 TOPS/W energy efficiency ¹⁶

Conductance-Dependent Output Scaling and ADC Reference Tuning

Conductance State Constraints and Mixed-Signal Interface Bottlenecks

While high-density analog computing assumes continuous conductance tuning, practical ReRAM fabrication limits non-volatile devices to low bit-precisions (typically 3-bit to 4-bit per cell, corresponding to 8 to 16 discrete conductance states) due to state variability, relaxation, and cycling degradation⁶. Furthermore, mapping high-precision neural network model parameters to restricted physical states causes variable current scaling across bitlines¹⁰.

When evaluating dot products, the total bitline current $I_{BL,j} = \sum_i V_i \cdot G_{i,j}$ can vary dynamically across several orders of magnitude depending on input activation density and column weight sums¹⁵. Converting this dynamic analog current back into a digital signal requires high-resolution Analog-to-Digital Converters (ADCs)². The signal path flows directly from the analog ReRAM crossbar core, which outputs dynamic analog currents into dynamic readout

interfaces equipped with configurable offset voltages (V_{offset}), adaptive comparison step sizes (ΔV), and thermal/drift compensation circuits to output digital quantized feature maps³. However, ADCs represent the primary bottleneck in AIMC macros, frequently consuming over 30% to 50% of total chip area and up to 70% of overall operational power².

ADC Architectures and Reference Tuning Methodologies

To optimize mixed-signal readout efficiency, several ADC architectures and calibration techniques are deployed at the crossbar periphery.

Successive Approximation Register (SAR) ADCs are widely utilized due to their compact layout and high energy efficiency at moderate resolutions (5 to 8 bits)³. Advanced SAR designs implement split-capacitance DACs and transconductance-enhanced dynamic comparators operating at high sampling rates (e.g., 70 MS/s)³. By utilizing bootstrapped switches and optimized capacitor topologies, dynamic range parameters such as Spurious-Free Dynamic Range (SFDR) and Signal-to-Noise and Distortion Ratio (SNDR) are maintained above 57 dB and 46 dB, respectively, occupying compact footprints ($< 50 \mu\text{m} \times 50 \mu\text{m}$)³.

Non-traditional readout circuits replace linear voltage/current sampling with time-based conversions³. Voltage-Controlled Oscillators (VCOs) or Current-Controlled Oscillators (CCOs) convert bitline currents into frequency-modulated pulse trains³. Integrated op-amp and current-mirror negative feedback loops maintain constant bitline bias while translating sum currents into digital counts via lightweight clock counters, dramatically reducing silicon area relative to array columns³.

Because memristive conductance distributions drift with temperature and electrical stress, static quantization reference levels introduce severe clipping and quantization noise¹⁹. Dynamic reference tuning addresses this issue by introducing adaptive reference voltages¹⁹. The transfer function mapping bitline current I_{BL} to digital code D_{out} is parameterized as:

$$D_{out} = \left\lfloor \frac{I_{BL} - I_{offset}}{\Delta I_{step}} \right\rfloor$$

By implementing on-chip programmable reference generators, I_{offset} , conversion step size ΔI_{step} , and target bitline bias V_{BL} are individually tuned on a per-module or per-column basis¹⁹. Temperature-independent ADC reference circuitry compensates for high-temperature conductance degradation, ensuring that system inference performance (e.g., 85.8% accuracy on CIFAR-10 at 120 °C) remains within 2% of ideal room-temperature targets¹⁹.

To bypass the mixed-signal conversion bottleneck entirely, novel architectures replace ADCs and digital activation units with Analog Content Addressable Memory (ACAM) arrays². Arbitrary non-linear activation functions (e.g., ReLU, Sigmoid) are converted into decision trees and mapped directly into programmable ACAM cells². The ACAM accepts the raw analog output current directly from the ReRAM crossbar and evaluates range-matching operations in the analog domain, outputting quantized digital class tokens and eliminating ADC power and area overheads².

Sneak-Path Calibration and Suppression in Passive Crossbars

The Sneak-Path Current Phenomenon

Passive crossbar arrays—configured in a zero-transistor-one-resistor (0T1R or 1R)

structure—offer the maximum theoretical integration density of $4F^2$ (where F is the minimum feature size) and enable multi-layer 3D stacking⁶. However, passive architectures lack individual isolation devices at each cell crosspoint¹². During read or write operations targeting a specific cell (i, j) , applied voltages inevitably create electrical potential differences across neighboring unselected and half-selected cells¹².

Unintended secondary currents—termed sneak-path currents—traverse adjacent low-resistance state (LRS) cells, bypassing the target cell and merging into the bitline output¹². In contrast to the target access path, where current flows directly from the selected wordline through the target device to the sensing bitline, sneak paths divert current through unselected wordlines and bitlines via adjacent LRS devices¹². These leakage currents re-enter the target bitline, degrading signal-to-noise ratio (SNR), introducing read/write crosstalk, causing incorrect state assessments, and severely limiting the maximum operational crossbar size¹².

Suppression Strategies and Active Feedback Topologies

Mitigating sneak-path currents requires suppressing the voltage differential across all unselected crosspoint devices²⁴.

Inserting an active selector element in series with the memristive device blocks low-voltage leakage currents¹². Active 1T1R (One Transistor, One ReRAM) structures insert a MOSFET selector to isolate individual cells¹⁴. While completely eliminating sneak paths and providing precise current compliance control, the large footprint of the access transistor ($12 -$) reduces the integration density of passive crossbars¹⁴. Alternatively, 1S1R (One Selector, One ReRAM) stacks and self-rectifying memristors integrate non-linear volatile threshold switches (such as ovonic threshold switches) or engineered Schottky/tunneling barriers into the device structure¹². These non-linear devices remain non-conductive at half-select voltages ($V_{DD}/2$) and conduct only when fully biased (V_{DD}), maintaining a $4F^2$ footprint¹².

In passive arrays lacking physical selectors, peripheral line drivers apply coordinated bias voltages across unselected lines²⁴. Under the $V_{DD}/2$ scheme, selected wordlines are driven to V_{read} and selected bitlines to 0 V , while all unselected wordlines and bitlines are biased at $V_{read}/2$ ²⁴. This forces the voltage drop across all unselected cells to either 0 V or $\pm V_{read}/2$, suppressing leakage current through non-linear memristors²⁴.

For analog vector-matrix accumulation, active feedback configurations employing operational amplifiers (Op-Amps) eliminate bitline sneak currents²⁶. The sensing bitlines are connected directly to the inverting input terminals of transimpedance Op-Amps, while the non-inverting

terminals are tied to ground (0 V)²⁶. Due to the virtual ground property of operational amplifiers, the potential of all active sensing bitlines is tightly regulated to 0 V ²⁷. By simultaneously driving all unselected wordlines to 0 V , the voltage drop across all unselected cells along the active bitline is held at exactly 0 V :

$$V_{cell,unselected} = V_{WL,unselected} - V_{BL,sensing} = 0\text{ V} - 0\text{ V} = 0\text{ V}$$

Consequently, sneak-path currents along the sensing path are zeroed out, allowing parallel current summation that accurately reflects target conductance states²⁶.

To minimize the circuit area occupied by per-column Op-Amps, multiport readout architectures measure cumulative array leakage and subtract sneak-path terms digitally or via differential analog loops²⁴. Conductance Compensation (CC) strategies exploit the intrinsic differential inputs of operational amplifiers to execute real-valued matrix operations without requiring dedicated analog inverters, simultaneously reducing pole counts, stabilizing feedback loops, and accelerating settling times²⁵.

Mitigation Framework	Architectural Implementation	Density / Area Footprint	Sneak Path Suppression Ratio	System Trade-offs
1T1R Active Array	MOSFET integrated at every cell node ¹⁴	Low ($12 -$) ¹⁴	Near 100% (Absolute isolation) ¹⁴	High area overhead; non-scalable 3D integration ¹⁴
1S1R / Self-Rectifying	Series threshold switch / engineered tunneling stack ¹²	Maximum ($4F^2$, 3D stackable) ⁶	High (Non-linearity dependent) ¹²	Complex material fabrication; limited operating voltage window ¹²
$V_{DD}/2$ Active Biasing	Dynamic peripheral driver voltage control ²⁴	Maximum ($4F^2$) ²⁴	Moderate (Requires non-linear $I -$	Standby power dissipation on half-selected lines ²⁴

			$V_{\text{cells}})^{24}$	
Op-Amp Virtual Grounding	Inverting Op-Amp sensing at bitline outputs ²⁶	High array density; peripheral area penalty ²⁶	High ($V_{\text{unselected}} \approx$) ²⁶	Requires one Op-Amp per column; prone to feedback loop bandwidth limits ²⁵
Conductance Compensation (CC)	Multiport differential Op-Amp feedback without inverters ²⁵	Compact peripheral area ²⁵	High (Mathematical / Circuit cancellation) ²⁴	Requires precise resistor array matching and differential balancing ²⁵

Architectural Trade-Offs and Hardware-Software Co-Design Outlook

Sustaining high computing fidelity in large-scale ReRAM AIMC macros requires balancing hardware physical constraints against software training paradigms¹. A holistic hardware-software co-design pipeline bridges high-level algorithmic abstractions—such as fast numerical sparse matrix solvers, weight-centric pre-distortion models, retraining-free batch normalization recalibration, and thermal-aware calibration layers—with physical circuit realizations¹. Hardware realizations incorporate differential SW-2T2R crosspoint arrays, on-chip dynamic ADC tuning circuits, virtual-ground transimpedance amplifiers, and table-based wear-leveling algorithms¹⁶. The interplay between parasitic wire resistance, readout precision, sneak-path currents, and lifetime degradation necessitates this unified design methodology¹⁵.

Compute Signal-to-Noise Ratio Optimization

The computational accuracy of an analog MVM engine is fundamentally bounded by the overall Compute Signal-to-Noise Ratio ($\text{SNR}_{\text{compute}}$)¹⁵. $\text{SNR}_{\text{compute}}$ is governed by a trade-off between thermal/conductance noise, parasitic IR drop attenuation, ADC clipping noise, and ADC quantization noise¹⁵:

$$\text{SNR}_{\text{compute}} = \frac{P_{\text{signal}}}{P_{\text{wire_IR}} + P_{\text{device_noise}} + P_{\text{quantization}} + P_{\text{clipping}}}$$

Optimizing $\text{SNR}_{\text{compute}}$ requires identifying an optimal sensing resistance (R_s^*) and bitline bias setting¹⁵. If sensing gain is set too high, column currents clip against the ADC full-scale

reference, introducing extreme non-linear clipping noise¹⁵. Conversely, if sensing gain is set too low, signal current collapses into the lower ADC quantization bins, degrading resolution¹⁵.

On-chip dynamic reference tuning adapts R_s^* and ADC thresholds in real-time, ensuring that the operational point sits precisely at the peak of the $SNR_{compute}$ curve across temperature and workload variations¹⁵.

Reliability and Wear-Leveling Interplay

A critical second-order consequence of compensating for IR drop is array endurance degradation¹⁸. To overcome spatial voltage drops during programming operations, write drivers

traditionally apply elevated write voltage pulses ($V_{write} + \Delta V_{comp}$) to ensure distant cells reach target conductance states¹⁸. However, proximal cells located near wordline drivers experience severe voltage stress, leading to over-SET and over-RESET conditions that accelerate dielectric breakdown and degrade cell endurance¹⁸.

Implementing crossbar-aware wear-leveling algorithms—such as Table-Based Crossbar Wear Leveling (XWL)—mitigates this localized stress¹⁸. By tracking cumulative runtime write stress per row and dynamically remapping heavily written weight matrices to less-stressed physical

regions, XWL improves total ReRAM array operational lifetime by over 324% with a minimal performance latency overhead (6.1%)¹⁸.

Synthesis and Future Outlook

Analog In-Memory Computing based on ReRAM crossbars provides a high-density, energy-efficient framework for accelerating matrix-vector computations¹. However, scaling these arrays to commercially viable sizes requires overcoming physical non-idealities, including parasitic IR drop, device I-V non-linearity, conductance-dependent output scaling, and sneak-path leakage⁷. Uncompensated crossbars suffer severe signal distortion, causing deep neural network classification accuracy to fall significantly below acceptable baselines⁷.

Addressing these challenges necessitates a multi-layered hardware-software co-design strategy¹. Fast computational modeling techniques—such as iterative refinement, super-nodal sparse LU factorization, and Weight-centric Prediction Models (WPM)—accelerate non-ideality

simulation by up to $15,000\times$ compared to conventional SPICE solvers while maintaining high numerical precision⁴. At the algorithmic and circuit levels, techniques such as retraining-free Batch Normalization recalibration, static and dynamic sparsification (BinSparX), peripheral current cancellation, and differential sign-weighted 2T2R topologies successfully recover classification performance to within 0.16% to 2% of ideal floating-point models¹.

Simultaneously, peripheral readout bottlenecks are mitigated through on-chip ADC reference tuning, time-based converters (VCO/CCO), and ADC-less Analog Content Addressable Memory (ACAM) engines, preserving signal fidelity across wide thermal and conductance ranges².

Parasitic sneak-path leakage in dense selectorless arrays is suppressed using active

virtual-ground transimpedance feedback and non-linear 1S1R stacks¹². By uniting pre-distortion algorithmic training with dynamic peripheral hardware calibration and crossbar wear-leveling, future ReRAM in-memory computing platforms can achieve high inference accuracy, superior energy efficiency ($> 40 \text{ TOPS/W}$), and extended operational lifetimes across edge and enterprise AI applications¹.

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